

Day 2 Plan

Maths: Ratio & proportion, partnership · Reasoning: Alphabet series · Science: Force & Newton's laws

Today's schedule (about 5½ hours with breaks)

#	Block	Time	What you do	Where
0	Day 1 recall	15 min	10 science questions from Day 1, no notes. Redo any Day 1 mistakes.	Page 2
1	Maths	75 min	5 simplification questions for speed → formulas & methods (10 min) → 2-3 book questions per pattern, timed → 10 Testbook PYQs on this topic → log mistakes	Section A · book · Testbook
-	Break	10 min	Away from the screen	-
2	Reasoning	45 min	Learn the method (10 min) → 15 timed questions, Testbook PYQs first → log	Section B · Testbook
-	Break	10 min	Away from the screen	-
3	Science	110 min	10 lessons with quick checks (≈55 min) → 20-question practice set → Testbook PYQs: force & laws of motion (S tier)	Section C · Testbook
-	Break	10 min	Away from the screen	-
4	General awareness	45 min	20 current-affairs MCQs (20 min) → static GA: Polity 2 — Fundamental Rights, DPSP, Fundamental Duties — Testbook GA PYQs on it (25 min)	CA PDF · Testbook
5	Close	15 min	Redo today's wrong questions · write 3 rules for tomorrow · send me your scores	Mistake log

Short day? Cut in this order: Testbook science PYQs → reasoning down to 10 questions → static GA down to 10 min. Never cut recall, maths or the science lessons. · Exam pace (average 54 sec a question): GA ~30 sec · Science ~45 · Reasoning ~50 · Maths ~65.

Block 0 — Day 1 recall (answers on the last page)

#	Question	Your answer
1	1 kWh = ? joules	
2	A knot is a unit of what?	
3	Convert 77 °F to °C	
4	What is the least count of a screw gauge?	
5	What is the dimension of power?	
6	Is pressure a scalar or a vector?	
7	Convert 72 km/h to m/s	
8	What does the area under a v-t graph give?	
9	A ball is thrown up at 20 m/s ($g = 10$). How long is it in the air in total?	
10	What is the average speed for equal distances covered at 30 km/h and 60 km/h?	

If you get fewer than 8 right, re-read only the Day 1 lessons you missed — not the whole chapter.

A · MATHS — Ratio & Proportion, Partnership

Core idea

A ratio **a : b** compares two quantities of the same kind. It works like 'parts': if $A : B = 2 : 3$, then out of every 5 parts, A has 2 and B has 3. So **A = 2/5 of the total** and **B = 3/5 of the total**. Most ratio questions become easy once you write each quantity as **(ratio number) × k**, where k is the size of one part.

Formulas

Concept	Formula / rule
Proportion	$a : b = c : d$ means $a/b = c/d \rightarrow \mathbf{ad = bc}$ (product of extremes = product of means)
Mean proportional of a, b	$\sqrt{ab}$
Third proportional of a, b	$\mathbf{b^2/a}$ ($a : b = b : x$)
Fourth proportional of a, b, c	$\mathbf{bc/a}$ ($a : b = c : x$)
Duplicate / sub-duplicate / triplicate	$a^2 : b^2 \cdot \sqrt{a} : \sqrt{b} \cdot a^3 : b^3$
Compound ratio of a:b and c:d	$ac : bd$
Componendo-dividendo	If $a/b = c/d$, then $\mathbf{(a + b)/(a - b) = (c + d)/(c - d)}$
Partnership	Profit share $\propto$ capital × time . Equal times $\rightarrow$ share $\propto$ capital only

Patterns — how to spot them, the method, and one solved example each

Pattern	Method	Solved example
1. Combine A:B and B:C	Make the common term (B) equal by multiplying each ratio	$A:B = 2:3$, $B:C = 4:5 \rightarrow \times 4$ and $\times 3 \rightarrow 8:12$ and $12:15 \rightarrow \mathbf{A:B:C = 8:12:15}$
2. Divide an amount	Total $\div$ sum of ratio parts = one part	₹1200 in 3:5 $\rightarrow 1200 \div 8 = 150 \rightarrow \mathbf{₹450 \text{ and } ₹750}$
3. Number added to reach a ratio	Set up $(a + x)/(b + x) = p/q$ and cross-multiply	Add x to 5 and 9 to get 2:3 $\rightarrow 3(5 + x) = 2(9 + x) \rightarrow \mathbf{x = 3}$ ($8:12 = 2:3 \checkmark$)
4. Mean / third / fourth proportional	Use the formulas above	Mean of 4, 9 = $\sqrt{36} = \mathbf{6}$ · third of 4, $6 = 36/4 = \mathbf{9}$ · fourth of 2, $3, 4 = 12/2 = \mathbf{6}$
5. Coins and notes	Convert the number ratio into a value ratio first	₹1, 50p, 25p coins in number ratio 1:2:4, total ₹60 $\rightarrow$ value ratio $1 : 1 : 1 \rightarrow ₹20$ each $\rightarrow \mathbf{20, 40, 80 \text{ coins}}$
6. Income–expenditure–savings	Income = ratio $\times$ x, expenditure = ratio $\times$ y; savings = income – expenditure	Incomes 5:3, expenditures 9:5, each saves ₹2000 $\rightarrow 5x - 9y = 2000$, $3x - 5y = 2000 \rightarrow x = 4000$, $y = 2000 \rightarrow \mathbf{\text{incomes ₹20,000 and ₹12,000}}$
7. Partnership	Multiply capital $\times$ months for each partner; split profit in that ratio	A: ₹6000 $\times$ 12 months, B: ₹8000 $\times$ 6 months $\rightarrow 72000 : 48000 = 3 : 2 \rightarrow$ profit ₹5000 $\rightarrow \mathbf{A ₹3000, B ₹2000}$

Traps

- If $A : B = 2 : 3$, A is **2/5 of the total**, not $2/3$. And A is $1/3$ **less** than B, while B is 50% **more** than A.
- Coins: the ratio in the question is usually by **number**. Convert it to a **value** ratio before dividing the money.
- Partnership: if a partner joins late or leaves early, use their **months**, not the full year.

B · REASONING — Alphabet & Alpha-numeric Series

Everything here depends on knowing letter positions **instantly**. Spend the first 5 minutes drilling the table below until you can answer 'what number is R?' in under 2 seconds.

Letter	A	B	C	D	E	F	G	H	I	J	K	L	M
Position	1	2	3	4	5	6	7	8	9	10	11	12	13
Reverse	26	25	24	23	22	21	20	19	18	17	16	15	14
Letter	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Position	14	15	16	17	18	19	20	21	22	23	24	25	26
Reverse	13	12	11	10	9	8	7	6	5	4	3	2	1

Trick	How it works
EJOTY	E = 5, J = 10, O = 15, T = 20, Y = 25. Jump to the nearest one and count ± 1 or ± 2 . (R = T - 2 = 18)
Reverse position	Position from Z = 27 - position . (Reverse of R = 27 - 18 = 9)
Opposite pairs (sum = 27)	AZ · BY · CX · DW · EV · FU · GT · HS · IR · JQ · KP · LO · MN. Memory hooks: BY , DeW , GT road, High School , Indian Railway , LO , MN
kth from the right	= (27 - k)th from the left
'n to the right of'	Add n to the position; 'n to the left of' → subtract n

Patterns, with solved examples

Pattern	Example	Solution
Fixed gap	B, E, H, K, ?	+3 each → N
Opposite pairs	AZ, BY, CX, ?	Next opposite pair → DW
Increasing / decreasing gap	Z, X, U, Q, ?	26, 24, 21, 17 → -2, -3, -4, -5 → 12 = L
Letter groups	ACE, GIK, MOQ, ?	Inside each group +2; group starts A, G, M (+6) → S → SUW
Alternating series	A, Z, C, X, E, V, ?	Two series: A, C, E, G (+2) and Z, X, V (-2) → next is G
Position-based	4th to the left of the 10th letter from the right?	10th from right = 27 - 10 = 17 (Q) → 17 - 4 = 13 → M
Letter-number mix	A1Z, B2Y, C3X, ?	Letter +1, number +1, opposite letter → D4W
Symbol sequences	'How many 3s are immediately preceded by 5?'	Scan left to right; underline each match. Read 'preceded' vs 'followed' carefully

Speed rule

Write the position numbers above the letters **before** looking for the pattern. Most letter series become simple number series (which you practised yesterday).

C · SCIENCE — Physics 2: Force & Laws of Motion

10 short lessons. Yesterday you learned how to **describe** motion. Today: **what causes** motion to start, stop or change. Answer every Quick check out loud before moving on.

Lesson 1 · What is a force? (about 4 min)

A **force** is a **push or a pull**. You can't see a force, only what it does. A force can:

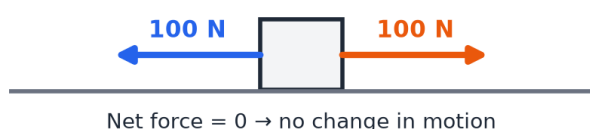
- make a still object move (kick a ball)
- stop a moving object (catch a ball)
- change its speed (press the accelerator)
- change its direction (hit a ball with a bat)
- change its shape (squeeze a sponge).

SI unit: **newton (N)**. Force is a **vector** — the direction matters (Day 1, Lesson 7).

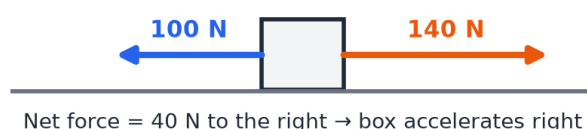
Contact forces (must touch)	Non-contact forces (act from a distance)
Muscular force, friction, normal force (a surface pushing back), tension in a rope, air resistance	Gravitational force, magnetic force, electrostatic force

Balanced vs unbalanced forces. Usually several forces act together. What matters is the **net (total) force**:

BALANCED: equal and opposite



UNBALANCED: one side stronger



- **Balanced** (net force = 0): no change in motion. A still object stays still; a moving one keeps the same speed and direction.
- **Unbalanced** (net force $\neq$ 0): the motion changes — the object speeds up, slows down or turns.

Quick check 1 — answer out loud before moving on (answers at the end)

a) Is gravity a contact or a non-contact force? b) 50 N pulls left and 80 N pulls right. What is the net force?

Lesson 2 · Newton's first law and inertia (about 6 min)

First law: an object stays **at rest**, or keeps **moving at the same speed in a straight line**, unless an **unbalanced force** acts on it.

In plain words: objects are 'lazy' — they keep doing whatever they were already doing. This laziness is called **inertia**: the tendency of an object to **resist any change** in its state of rest or motion. That's why the first law is also called the **law of inertia**. (The idea came from Galileo; Newton stated it in his book Principia, 1687.)

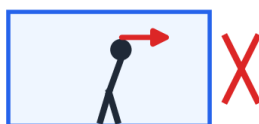
Mass is the measure of inertia. More mass → harder to start, stop or turn. Pushing a truck is harder than pushing a bicycle, even though both are at rest. Inertia does **not** depend on speed, shape or volume — only on mass.

Bus STARTS suddenly
Inertia of rest



Feet move with the bus;
upper body stays at rest
→ you fall BACKWARD

Bus STOPS suddenly
Inertia of motion



Feet stop with the bus;
upper body keeps moving
→ you fall FORWARD

Bus TURNS left
Inertia of direction



Body tries to keep going
straight → you lean to
the RIGHT (outward)

Type of inertia	Meaning	Everyday examples (frequent questions)
Inertia of rest	A still object resists starting to move	Falling backward when a bus starts · dust leaves a carpet when it's beaten · a coin on a card drops into the glass when the card is flicked away
Inertia of motion	A moving object resists stopping	Falling forward when a bus brakes · an athlete runs before a long jump · seat belts hold you when a car stops suddenly · tightening a hammer head by striking the handle on the ground
Inertia of direction	A moving object resists turning	Leaning outward when a bus turns · mud flies off a spinning wheel along the tangent · sparks fly off a grinding wheel along the tangent

Quick check 2 — answer out loud before moving on (answers at the end)

a) Which has more inertia: a truck or a bicycle? b) Dust leaves a carpet when it's beaten. Which type of inertia is this?

Lesson 3 · Momentum — the 'quantity of motion' (about 3 min)

Why is it harder to stop a loaded truck than a bicycle at the same speed? And why does a bullet hurt even though it's tiny? Because stopping something depends on **both its mass and its velocity**. Together they make **momentum**:

Momentum $p = \text{mass} \times \text{velocity} = m \times v$ · SI unit: **kg·m/s** · it's a **vector** (same direction as the velocity).

Worked example — momentum

A 0.15 kg cricket ball at 30 m/s: $p = 0.15 \times 30 = 4.5 \text{ kg·m/s}$.

A 1500 kg car at 20 m/s: $p = 1500 \times 20 = 30,000 \text{ kg·m/s}$.

A car at rest: $v = 0$, so $p = 0$. A heavy object at rest has no momentum.

Quick check 3 — answer out loud before moving on (answers at the end)

a) What is the momentum of a 2 kg ball moving at 5 m/s? b) Can a stationary object have momentum?

Lesson 4 · Newton's second law: $F = ma$ (about 6 min)

Second law: the force on an object equals the **rate of change of its momentum**. For a fixed mass this becomes the most useful formula in mechanics:

$$F = m \times a \text{ (force = mass} \times \text{acceleration)}$$

Read it both ways: • the **same force** gives a light object a big acceleration and a heavy object a small one; • to give the **same acceleration**, a heavier object needs a bigger force.

Definition of 1 newton: the force that gives a 1 kg mass an acceleration of 1 m/s^2 . So $1 \text{ N} = 1 \text{ kg} \cdot \text{m/s}^2$. **1 kilogram-force (kgf)** = the weight of 1 kg = $1 \times 9.8 = 9.8 \text{ N}$.

Worked example — $F = ma$

(1) 20 N acts on a 4 kg block: $a = F/m = 20/4 = 5 \text{ m/s}^2$.

(2) A 1200 kg car goes from 0 to 20 m/s in 10 s: $a = 20/10 = 2 \text{ m/s}^2 \rightarrow F = 1200 \times 2 = 2400 \text{ N}$.

(3) Force to stop a 1000 kg car moving at 20 m/s in 4 s: $a = 20/4 = 5 \text{ m/s}^2 \rightarrow F = 1000 \times 5 = 5000 \text{ N}$.

How the three laws fit together: the 1st law says **what** a force does (changes motion), the 2nd says **how much** ($F = ma$), and the 3rd says forces always come in **pairs**.

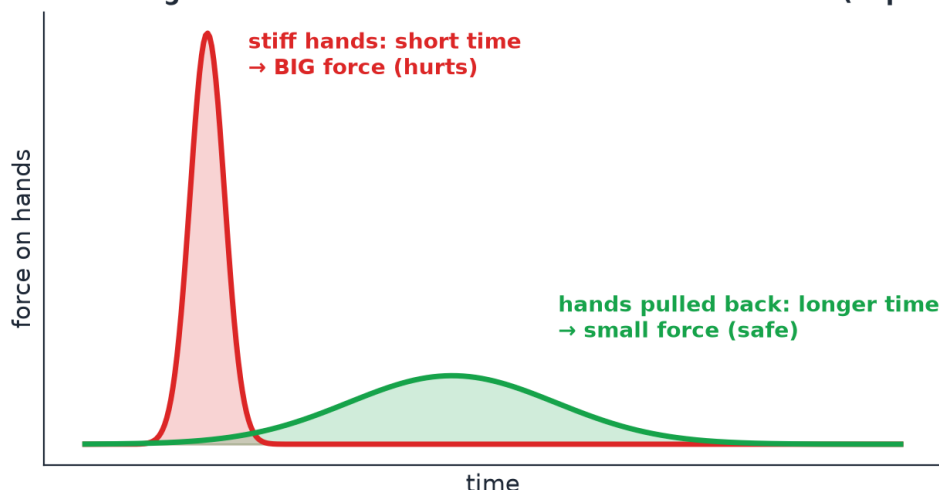
Quick check 4 — answer out loud before moving on (answers at the end)

a) 10 N acts on a 2 kg mass. What is its acceleration? b) What force gives a 2 kg mass an acceleration of 3 m/s^2 ? c) $1 \text{ kgf} = ? \text{ N}$

Lesson 5 · Impulse — why 'giving it more time' reduces the force (about 5 min)

Rewrite the second law: $F = (\text{change in momentum}) \div \text{time}$. So for the **same change in momentum**, if the stopping **time is longer, the force is smaller**. The product $F \times t$ is called **impulse**, and it equals the change in momentum. Unit: $\text{N} \cdot \text{s}$ (the same as $\text{kg} \cdot \text{m/s}$).

Same change in momentum = same AREA under both curves (impulse)



Worked example — catching a ball

A 0.15 kg ball at 20 m/s is caught and stopped. Change in momentum = $0.15 \times 20 = 3 \text{ kg} \cdot \text{m/s}$.

Stiff hands, stopped in 0.01 s: $F = 3 \div 0.01 = 300 \text{ N}$ (it stings).

Hands pulled back, stopped in 0.1 s: $F = 3 \div 0.1 = 30 \text{ N}$ — ten times less.

Situation	How the time is made longer
Cricketer catching a ball	Pulls the hands back while catching
High jump / long jump	Athletes land on foam beds or a sand pit
Car safety	Airbags, crumple zones, and shock absorbers in vehicles
Packing glassware	Straw, thermocol or bubble wrap around fragile items
Jumping from a height	Bending the knees on landing

Quick check 5 — answer out loud before moving on (answers at the end)

Why do high jumpers land on foam instead of a hard floor? Use the word 'time' in your answer.

Lesson 6 · Newton's third law — action and reaction (about 5 min)

Third law: to every action there is an **equal and opposite reaction**. When body A pushes body B, body B pushes back on A with the **same size** of force, in the **opposite direction**, at the **same instant**.

Why don't they cancel out? Because they act on **different bodies**. Forces only cancel when they act on the **same** body. When you walk, your foot pushes the **ground** backward; the ground pushes **you** forward — and you move.

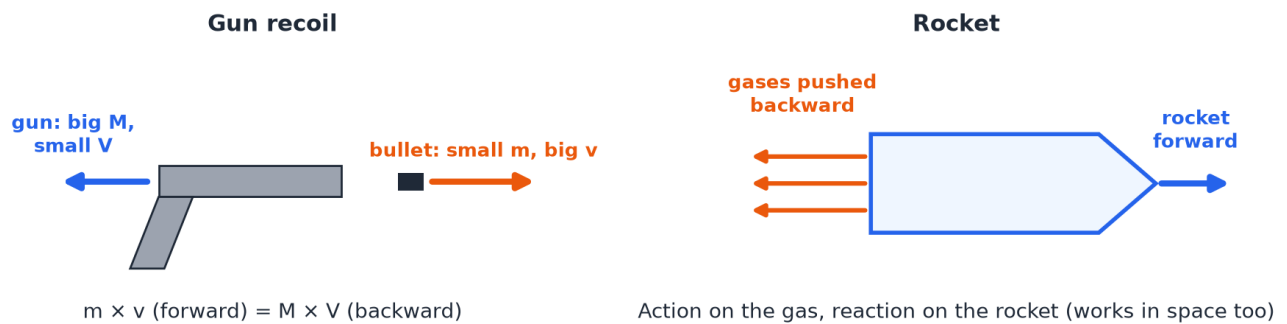
Example	Action (by ...)	Reaction (on ...)
Walking	Foot pushes the ground backward	Ground pushes you forward
Swimming	Hands push water backward	Water pushes the swimmer forward
Rocket	Rocket pushes hot gas backward	Gas pushes the rocket forward (works in empty space too)
Gun firing	Gun pushes the bullet forward	Bullet pushes the gun backward (recoil)
Jumping off a boat	You push the boat backward	The boat pushes you forward — the boat moves away
Bird flying	Wings push air downward	Air pushes the bird upward

Quick check 6 — answer out loud before moving on (answers at the end)

a) When you walk, which way do you push the ground, and which way does it push you? b) Why don't action and reaction cancel each other?

Lesson 7 · Conservation of momentum (about 5 min)

If no outside force acts on a group of objects, their **total momentum stays the same** — before and after any collision or explosion. This follows directly from the third law: the two bodies get equal and opposite pushes for the same time, so one gains exactly the momentum the other loses.



Worked example — gun recoil

A 4 kg gun fires a 0.02 kg (20 g) bullet at 400 m/s. Before firing, the total momentum = 0 (both at rest).

After: bullet momentum forward = $0.02 \times 400 = 8 \text{ kg}\cdot\text{m/s}$. So the gun must have $8 \text{ kg}\cdot\text{m/s}$ backward.

Recoil velocity $V = 8 \div 4 = \mathbf{2 \text{ m/s}}$ backward. The heavy gun moves slowly; the light bullet moves fast.

Worked example — jumping off a boat

A 50 kg person jumps from a 100 kg boat at 2 m/s. Person's momentum = $100 \text{ kg}\cdot\text{m/s}$ forward.

Boat: $100 \times V = 100 \rightarrow \mathbf{V = 1 \text{ m/s}}$ backward.

Formula: $m_1v_1 = m_2v_2$ (when everything starts from rest). A **rocket** works on conservation of momentum: the gas gets momentum backward, so the rocket gets equal momentum forward.

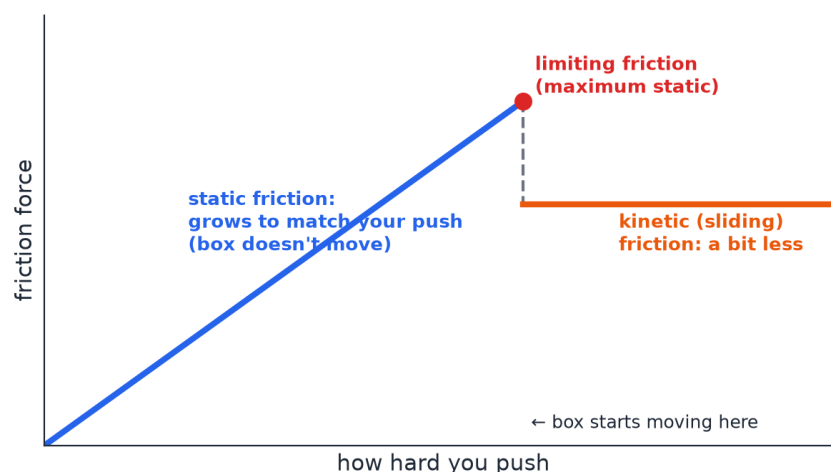
Quick check 7 — answer out loud before moving on (answers at the end)

A 50 g bullet is fired at 400 m/s from a 4 kg gun. What is the gun's recoil velocity?

Lesson 8 · Friction (about 6 min)

Friction is the force that **opposes relative motion** between two surfaces in contact. It happens because even smooth-looking surfaces have tiny bumps that **interlock**. Friction always acts **opposite** to the direction of motion (or of the attempted motion).

What happens when you push a heavy box:



- At first the box doesn't move: **static friction** exactly matches your push, growing as you push harder.

- Static friction has a maximum, called **limiting friction**. Push beyond it and the box starts to slide.
- Once it's moving, **kinetic (sliding) friction** is a little smaller — that's why it's easier to **keep** a box moving than to start it.
- **Rolling friction** is much smaller still — that's why wheels were such an important invention.

Order (largest → smallest): static (limiting) > sliding > rolling.

Worked example — will the box move?

A 10 kg box on a floor, coefficient of friction $\mu = 0.4$, $g = 10$. The floor pushes up with $N = mg = 100$ N.

Maximum (limiting) friction $= \mu N = 0.4 \times 100 = 40$ N.

Push with 30 N $\rightarrow 30 < 40$, so the box **doesn't move**; friction = 30 N (it only matches the push).

Push with 50 N $\rightarrow$ it slides.

Increase friction (when we need grip)	Reduce friction (when it wastes energy)
Treads on tyres and shoe soles · spikes on athletes' shoes · sand on icy or slippery roads · chalk powder on gymnasts' hands	Lubricants (oil, grease) · ball bearings (turn sliding into rolling) · polishing surfaces · streamlined shapes for cars, planes and fish (less fluid friction, called drag)

Friction is a 'necessary evil': without it we couldn't walk, write, or strike a matchstick (friction produces heat). But it also wears out machine parts and wastes energy as heat.

Exam trap

- Friction does **not** depend (approximately) on the **area of contact**. It depends on the **nature of the surfaces** and the **normal force** (how hard they're pressed together).
- Fluid friction (drag) **does** depend on speed and shape — that's why streamlining helps.

Quick check 8 — answer out loud before moving on (answers at the end)

a) Put static, rolling and sliding friction in order, largest to smallest. b) Why do tyres have treads?

Lesson 9 · Circular motion and centripetal force (about 6 min)

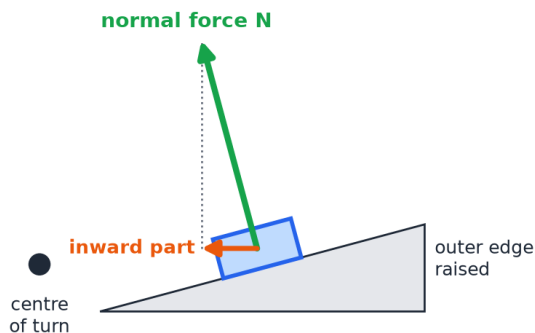
From Day 1: moving in a circle means the **direction keeps changing**, so it is **accelerated motion**. By the second law, an acceleration needs a force. In a circle, that force must point **towards the centre**. It's called the **centripetal force** ('centre-seeking'):

$$F = m v^2 / r \quad (m = \text{mass}, v = \text{speed}, r = \text{radius of the circle})$$

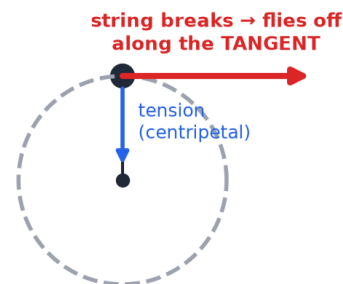
Centripetal force is **not a new kind of force**. It's a job done by some existing force:

Situation	Which force pulls towards the centre
Stone whirled on a string	Tension in the string
Moon going round the Earth, satellites	Gravity
Car turning on a flat road	Friction between the tyres and the road
Car on a banked (tilted) road	Part of the road's normal force
Electron around the nucleus	Electrostatic attraction

Banked road: the tilt gives the inward (centripetal) push



Whirling stone



- **Banking of roads:** the outer edge of a curve is raised, so part of the road's push points towards the centre. Vehicles can then turn safely without relying only on friction (important on wet roads).
- A **cyclist leans inward** on a turn for the same reason.
- If the string breaks, the stone flies off along the **tangent** — not outward — because of **inertia of direction**.

Worked example — centripetal force

A 1000 kg car takes a turn of radius 100 m at 20 m/s: $F = 1000 \times 20^2 \div 100 = \mathbf{4000\text{ N}}$.

At double the speed (40 m/s): $F = 1000 \times 1600 \div 100 = 16,000\text{ N} = \mathbf{4\text{ times}}$, because F depends on v^2 . That's why speeding on curves is dangerous.

Centrifugal force is the **apparent outward** pull you feel inside a turning vehicle. It isn't a real force acting on you from outside — it's your inertia. It is still used in machines that spin things fast: **cream separators**, **washing-machine dryers** and laboratory **centrifuges**.

Quick check 9 — answer out loud before moving on (answers at the end)

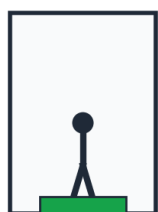
a) An 800 kg car turns on a 50 m radius at 10 m/s. What is the centripetal force? b) Why does a cyclist lean inward on a turn?

Lesson 10 · Apparent weight in a lift (about 4 min)

A weighing machine doesn't measure your weight directly. It measures how hard it has to **push up** on you (the normal force R). In a lift that's accelerating, that push changes:

At rest / constant speed

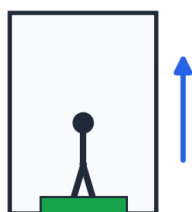
$$R = mg$$



normal weight

Accelerating UP

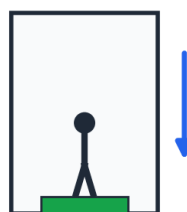
$$R = m(g + a)$$



feels **HEAVIER**

Accelerating DOWN

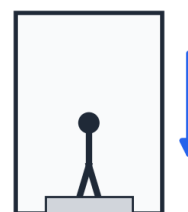
$$R = m(g - a)$$



feels **LIGHTER**

Free fall (cable cut)

$$R = 0$$



WEIGHTLESS

R = reading on the weighing machine (normal force) · blue arrow = direction of the lift's acceleration

Worked example — 60 kg person, $g = 10$, $a = 2 \text{ m/s}^2$

At rest: $R = 60 \times 10 = 600 \text{ N}$ (reads 60 kg).

Accelerating up: $R = 60 \times (10 + 2) = \mathbf{720 \text{ N}}$ (reads 72 kg — feels heavier).

Accelerating down: $R = 60 \times (10 - 2) = \mathbf{480 \text{ N}}$ (reads 48 kg — feels lighter).

Free fall ($a = g$): $R = 60 \times (10 - 10) = \mathbf{0}$ → weightless. Astronauts in orbit feel weightless for the same reason: they're in continuous free fall.

Exam trap

Moving at **constant speed** (up or down) = no acceleration → the reading is **normal**. Only **acceleration** changes it.

Quick check 10 — answer out loud before moving on (answers at the end)

A 50 kg person is in a lift accelerating downward at 2 m/s^2 ($g = 10$). What does the weighing machine read in newtons?

Science 1-page summary — revise from this tomorrow

Topic	Must remember
Force	Push or pull · unit N · vector · balanced (net 0) → no change; unbalanced → motion changes
First law	Law of inertia · mass = measure of inertia · rest / motion / direction examples (bus, carpet, athlete, mud off a wheel)
Momentum	$p = mv$ · kg·m/s · vector · zero at rest
Second law	$F = \text{rate of change of momentum} = ma$ · $1 \text{ N} = 1 \text{ kg} \cdot \text{m/s}^2$ · $1 \text{ kgf} = 9.8 \text{ N}$
Impulse	$F \times t = \text{change in momentum}$ · more time → less force (catching, airbags, sand pit, bending the knees)
Third law	Equal and opposite, on DIFFERENT bodies, so they don't cancel · walking, swimming, rocket, recoil
Conservation	Total momentum is constant without an external force · recoil: $m_1v_1 = m_2v_2$ · rocket
Friction	Opposes relative motion · static > sliding > rolling · limiting friction = μN · independent of contact area · ball bearings, lubricants
Circular motion	Centripetal $F = mv^2/r$ towards the centre · double the speed → 4× the force · banking · string breaks → tangent
Centrifugal	Apparent outward force (inertia) · cream separator, washing-machine dryer, centrifuge
Lift	Up: $m(g + a)$ heavier · down: $m(g - a)$ lighter · free fall: 0 weightless · constant speed: normal

Science practice set — 20 questions · target 6 minutes

#	Question
1	Newton's first law is also called the: (a) law of momentum (b) law of inertia (c) law of action-reaction (d) law of gravitation
2	The inertia of a body depends on its: (a) mass (b) shape (c) volume (d) speed
3	Passengers fall forward when a bus stops suddenly because of: (a) inertia of direction (b) friction (c) inertia of rest (d) inertia of motion
4	The SI unit of momentum is: (a) J (b) N (c) kg·m/s (d) kg·m/s ²
5	A 20 N force acts on a 4 kg mass. Its acceleration is: (a) 80 m/s ² (b) 5 m/s ² (c) 0.2 m/s ² (d) 24 m/s ²
6	A rocket works on the principle of conservation of: (a) momentum (b) mass (c) charge (d) energy
7	A cricketer pulls the hands back while catching to: (a) increase the force (b) increase the ball's speed (c) reduce the ball's momentum (d) increase the time of impact and reduce the force
8	Action and reaction forces: (a) are unequal (b) act on the same body (c) act on different bodies (d) cancel each other
9	1 kgf is approximately equal to: (a) 1 N (b) 9.8 N (c) 98 N (d) 0.98 N
10	A 0.02 kg bullet is fired at 500 m/s from a 5 kg gun. The recoil velocity of the gun is: (a) 2 m/s (b) 5 m/s (c) 10 m/s (d) 1 m/s
11	Which friction is the smallest? (a) sliding (b) rolling (c) static (d) limiting
12	Ball bearings are used to: (a) reduce weight (b) increase friction (c) convert sliding friction into rolling friction (d) increase speed
13	Friction between two solid surfaces is (approximately) independent of: (a) the nature of the surfaces (b) the normal force (c) the area of contact (d) roughness
14	Roads are banked on curves mainly to: (a) reduce the weight of vehicles (b) increase friction (c) increase the speed limit (d) provide centripetal force
15	A stone whirled on a string is released when the string breaks. It moves: (a) along the tangent (b) in a spiral (c) towards the centre (d) radially outward
16	If a car's speed on the same curve is doubled, the centripetal force needed becomes: (a) the same (b) double (c) four times (d) half
17	A person in a lift accelerating upward feels: (a) lighter (b) heavier (c) weightless (d) no change
18	The apparent weight of a person in a freely falling lift is: (a) 2mg (b) zero (c) infinite (d) mg
19	The momentum of a 1500 kg car moving at 20 m/s is: (a) 7500 (b) 3,00,000 kg·m/s (c) 3000 (d) 30,000
20	The force needed to stop a 1000 kg car moving at 20 m/s in 4 s is: (a) 500 N (b) 2000 N (c) 5000 N (d) 80,000 N

Score	Time taken	Wrong Q numbers	Mistake type (concept / formula / calc / careless / recall)
/ 20			

Answers

Science practice set

#	Ans	Why / lesson to re-read
1	b	Lesson 2
2	a	Mass is the measure of inertia (Lesson 2)
3	d	Upper body keeps moving (Lesson 2)
4	c	$p = mv \rightarrow \text{kg}\cdot\text{m/s}$ (Lesson 3)
5	b	$a = 20/4$ (Lesson 4)
6	a	Lesson 7
7	d	Impulse: more time, less force (Lesson 5)
8	c	That's why they don't cancel (Lesson 6)
9	b	Lesson 4
10	a	$0.02 \times 500 = 10 \rightarrow 10/5 = 2$ (Lesson 7)
11	b	Lesson 8
12	c	Lesson 8
13	c	Lesson 8 trap
14	d	Lesson 9
15	a	Inertia of direction (Lesson 9)
16	c	$F \propto v^2$ (Lesson 9)
17	b	$R = m(g + a)$ (Lesson 10)
18	b	Lesson 10
19	d	1500×20 (Lesson 3)
20	c	$a = 5 \text{ m/s}^2$, $F = 1000 \times 5$ (Lesson 4)

Quick checks

QC	Answers
1	a) non-contact b) 30 N to the right
2	a) the truck (more mass) b) inertia of rest
3	a) $2 \times 5 = 10 \text{ kg}\cdot\text{m/s}$ b) no — $v = 0$, so $p = 0$
4	a) 5 m/s^2 b) 6 N c) 9.8 N
5	Foam makes the stopping time longer, so for the same change in momentum the force on the body is smaller
6	a) you push the ground backward; it pushes you forward b) they act on different bodies
7	$0.05 \times 400 = 20 \text{ kg}\cdot\text{m/s} \rightarrow V = 20/4 = 5 \text{ m/s}$ backward
8	a) static > sliding > rolling b) to increase friction (grip)
9	a) $800 \times 10^2 \div 50 = 1600 \text{ N}$ b) leaning provides the inward (centripetal) force needed to turn
10	$R = 50 \times (10 - 2) = 400 \text{ N}$

Day 1 recall (Block 0)

#	Answer	#	Answer
1	$3.6 \times 10^6 \text{ J}$	6	Scalar
2	Speed	7	20 m/s
3	$(77 - 32) \times 5/9 = 25 \text{ }^\circ\text{C}$	8	Displacement
4	0.01 mm	9	$2u/g = 4 \text{ s}$
5	$[\text{M L}^2 \text{ T}^{-3}]$	10	$2 \times 30 \times 60 \div 90 = 40 \text{ km/h}$

Send me tonight

- Day 1 recall score /10 · Science practice score /20 and time · Maths and Reasoning: questions attempted, correct, average seconds per question · the numbers you got wrong. I'll add the mistakes to your log and pick tomorrow's recall questions from them.